

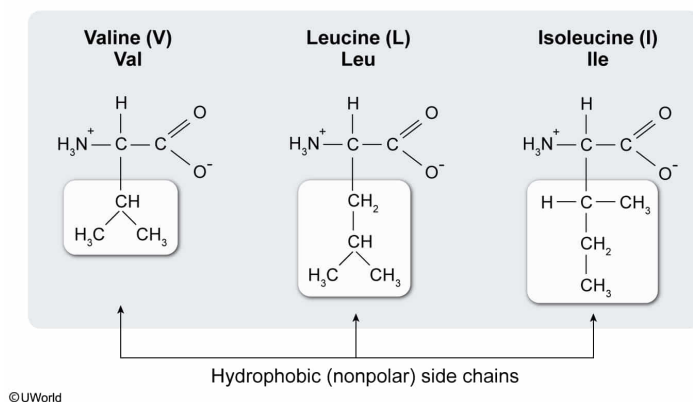
The amino acids Val, Leu, and Ile are all degraded outside the liver by an aminotransferase and an α -keto acid dehydrogenase complex. These amino acids can all be classified as:

- ☐ A. acidic. [1%]
- ☐ B. basic. [1%]
- ✔ ☒ C. hydrophobic. [93%]
- ☐ D. hydrophilic. [3%]

Correct

93% answered correctly

Explanation:



Amino acids have three components: an amino group, a carboxyl group, and a **side chain**, which are each bonded to the same carbon atom (known as the α -carbon). The various amino acid structures differ only by the composition of the substituent side chains, which contain different **functional groups** that give amino acids specific **characteristics**.

Amino acids are classified by the characteristics of their side chains. Major **classifications** include charged (acidic or basic), **hydrophobic** (nonpolar), and hydrophilic (polar and charged). Some amino acids fall into **more than one** of these categories. For example, the charged amino acid side chains are also hydrophilic.

Val, Leu, and Ile are the three-letter abbreviations for valine, leucine,

and isoleucine, respectively. The side chains of these amino acids are similar in functionality. Each is a hydrocarbon chain of varying length with branching at some point in the side chain. Because the side chains of these amino acids are composed of **nonpolar** hydrocarbon chains, they are classified as hydrophobic.

(Choices A and B) **Acidic** and **basic** amino acids contain negatively and positively charged side chains, respectively, at physiological pH, and they have a nonzero net charge. In contrast, valine, leucine, and isoleucine contain uncharged side chains and have a net charge of zero at physiological pH.

(Choice D) The side chains of valine, leucine, and isoleucine do not contain any **polar functional groups**. The hydrocarbon side chains are nonpolar, making these amino acids **hydrophobic** rather than hydrophilic.

Educational objective:

Amino acids are made up of an amino group, a carboxyl group, and a side chain, all bonded to the same carbon atom. The side chains contain different functional groups that give amino acids specific characteristics, which allow the classification of amino acids based on their properties. Four main classifications of amino acids include acidic, basic, hydrophilic, and hydrophobic.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions